

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 04/15/2020

Quiz 10: Section 4.4, 4.5, 4.6

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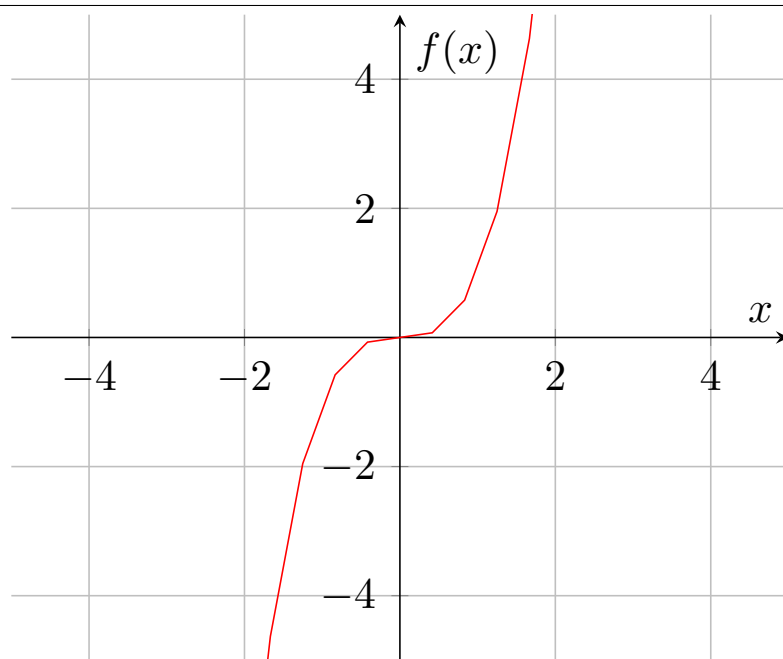
Section 4.4

3 pts

1. Sketch a possible graph of a function which satisfies

$$f' > 0, f'' < 0 \text{ on } (-\infty, 0), \quad f' > 0, f'' > 0 \text{ on } (0, \infty)$$

First, we will use the information above to determine the behavior of f and its graph. On the interval $(-\infty, 0)$, f is increasing and concave down. On the interval $(0, \infty)$, f is increasing and concave up. Following this, we can sketch a possible graph of f . Notice that derivative information is not sufficient to determine the y -coordinates of points on the curve given in the graph.



Section 4.5

3 pts

1. Suppose an airline policy states that all baggage must be box-shaped with a sum of length, width, and height not exceeding 64 inches. What are the dimensions a square-based box with the greatest volume under these conditions?
- (A) The box should have a base with sides that are 17 inches in length and a height of 30 inches.
- (B) The box should have a base with sides that are $64/2$ inches in length and a height of $64/2$ inches.
- (C) The box should have a base with sides that are $64/3$ inches in length and a height of $64/3$ inches.**
- (D) The box should have a base with sides that are 30 inches in length and a height of 2 inches.

We consider square-based box whose length and width are w and whose height is h . By the airline policy, the constraint is $2w + h = 64$. The objective function is the volume, $V = w^2h$. Either w or h may be eliminated from the objective function; the constraint $h = 64 - 2w$ implies that the volume is

$$V = w^2h = w^2(64 - 2w) = 64w^2 - 2w^3.$$

The objective function has now been expressed in terms of a single variable. Notice that w is nonnegative and cannot exceed 32, so the domain of V is $0 \leq w \leq 32$. The critical points satisfy

$$V'(w) = 128w - 6w^2 = 2w(64 - 3w) = 0,$$

which has roots $w = 0$ and $w = 64/3$. By the First (or Second) Derivative Test, $w = 64/3$ corresponds to a local maximum. At the endpoints, $V(0) = V(32) = 0$. Therefore, the volume function has an absolute maximum of

$$V(64/3) \approx 9709 \text{ in}^3.$$

The dimensions of the optimal box are $w = 64/3$ in and $h = 64 - 2w = 64/3$ in, so the optimal box is a cube.

Section 4.6

2 pts

1. Find the best linear approximation to $\sin(x)$ at $x = 0$ and use it to approximate $\sin(\pi/72)$.

- (A) The best linear approximation is $L(x) = x^2$ and $\sin(\pi/72) \approx (\pi/72)^2$.
- (B) The best linear approximation is $L(x) = -x$ and $\sin(\pi/72) \approx -\pi/72$.
- (C) The best linear approximation is $L(x) = x + 1$ and $\sin(\pi/72) \approx \pi/72 + 1$.
- (D) The best linear approximation is $L(x) = x$. So, $\sin(\pi/72) \approx \pi/72$.**

We first construct a linear approximation $L(x) = f(a) + f'(a)(x - a)$, where $f(x) = \sin(x)$ and $a = 0$. Noting that $f(0) = 0$ and $f'(0) = \cos(0) = 1$, we have

$$L(x) = 0 + 1(x - 0) = x.$$

So, our linear approximation yields the estimate

$$\sin(\pi/72) \approx L(\pi/72) = \pi/72 \approx 0.04363.$$

A calculator gives $\sin(\pi/72) \approx 0.04362$, so the approximation is accurate to four decimal places.

2 pts

2. Find the linear approximation to $f(x) = \sqrt[3]{x}$ at $x = 1$. Is this approximation an overestimate or underestimate? Explain why **without computing the error** of your approximation.

- (A) The best linear approximation is $L(x) = \frac{1}{3}x + \frac{2}{3}$. Since $f''(1) < 0$, we know this is an overestimate, as the graph of $f(x)$ lies below the graph of $L(x)$
- (B) The best linear approximation is $L(x) = \frac{1}{3}x + \frac{2}{3}$. Since $f''(1) < 0$, we know this is an underestimate, as the graph of $f(x)$ lies above the graph of $L(x)$
- (C) The best linear approximation is $L(x) = (3x)^{-2/3}$. Since $f'(x) = L(x)$, we know this is an overestimate, as the linear approximation coincides with the derivative.
- (D) The best linear approximation is $L(x) = (3x)^{-2/3}$. Since $f'(x) = L(x)$, we know this is an underestimate, as the linear approximation coincides with the derivative.

First, we find the linear approximation to $f(x) = \sqrt[3]{x}$ at $x = 1$. Note that $f(1) = 1$, and observe that $f'(x) = (3x)^{-2/3}$. So, $f'(1) = 1/3$, and the linear approximation at $x = 1$, denoted L_1 , is given by

$$L_1(x) = 1 + \frac{1}{3}(x - 1) = \frac{1}{3}x + \frac{2}{3}.$$

Using this results, we now find that

$$\sqrt[3]{2} \approx L_1(2) = \frac{4}{3} \approx 1.333.$$

As $f''(x) = -(2/9)x^{-5/3}$, we see that $f''(x) < 0$ for $x > 0$. Correspondingly, f is concave down at $x = 1$. Hence, L_1 lies above the graph of f and the approximation is an overestimate.